

Fatih Gurbuz

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Hiring Committee

Department of Robotics

University of Michigan

Dear Hiring Committee,

My name is Fatih. I want to build systems that make complex technology easier for people to use. That goal began when I worked on privacy-preserving genomics research supported by NIH and NSF projects, where I designed secure, automated workflows for large-scale biomedical data. I learned how careful software design can make advanced tools more transparent and accessible for the people who rely on them.

The Applications Programmer Associate position in the Robotics Department fits that same mission. I am excited to contribute to a team that develops, supports, and improves software used by researchers and students pushing the boundaries of robotics. I can bring experience building production-grade applications in JavaScript and Python, writing and optimizing database queries, and debugging and maintaining systems in active environments. At Amazon Lab126 I worked on embedded multimodal Alexa systems that required reliability and performance at scale. At bitHuman.ai I developed real-time web interfaces and backend APIs that powered interactive AI applications, and at romorom I built and managed Python based data pipelines that supported users in more than 200 regions. Across these projects I learned fast from scratch, iterated quickly, and stayed close to user needs.

I enjoy writing clear, well-documented code, collaborating with non-technical stakeholders, and maintaining security and compliance standards. I also value environments like Michigan Engineering where technology and education meet, and where innovation supports a larger public mission.

I plan to pursue graduate study and eventually a PhD focused on computing. Joining Michigan's Robotics Department would allow me to grow as a software engineer while contributing to systems that support research and learning. I have attached my resume and would be grateful for the opportunity to discuss how I can help your team.

Thank you for your time and consideration.

Regards,

Fatih Gurbuz

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gurbuz1gurbuz@outlook.com • +1 216 417 9302 • U.S. Citizen

EDUCATION

Computer Science, Bachelor's	Case Western Reserve University	May 2024
- Economics, <i>Minor</i> - Emerging Leaders Program, Design for America, Entrepreneurship Club, Mediterranean Cultural Club (co-founder) - GPA: 3.7		

LEADERSHIP

Co-Founder	LabScoutr (Nonprofit)	September 2025 –
- Co-founded interactive platform connecting researchers with faculty mentors, addressing inefficiencies in academic research discovery and reducing barriers to entry for students - Architected web application using React, Google Cloud, MongoDB, Neo4j with automated data pipelines processing research metrics from 100+ universities including publication data, citation analytics, and grant funding status - Developing AI-powered lab matching system analyzing student skills against research requirements using NLP and machine learning to provide personalized recommendations and skill gap analysis - Technologies: React, TypeScript, Google Cloud Suite, Python, Neo4j		
Founder	romorom	May 2024 –
- Developed an on-demand, borderless cellular connectivity platform providing service in 201 countries with a current user base across 28 countries - Built cloud-native microservices architecture using Firebase Cloud Functions with automated secure eSIM assignment, Stripe payment and webhook event processing, real-time balance management, and atomic transactions - Built automated data processing pipelines using Python (Pandas, NumPy) to clean, normalize, and validate data from thousands of providers across 200+ countries - Developed ETL workflows to standardize disparate data formats, handle missing values, and ensure data quality for provider matching and rate optimization - Implemented intelligent balance caching system reducing third-party API calls by 80% while maintaining real-time updates - Built responsive email notification system for user communications - Built Flutter mobile applications for iOS and Android with in-app payments and native performance optimization - Built comprehensive user management system with Firebase Authentication, account settings, and transaction history - Implemented advanced device detection and platform-specific UI adaptations for optimal user experience - Technologies: Flutter, Dart, Node.js, Firebase (Cloud Functions, Firestore, Authentication, Analytics), React, Stripe API, Google Cloud Platform, Intercom SDK, iOS Core Telephony APIs		

RESEARCH EXPERIENCE

Research Assistant, Cleveland, OH	Case Western Reserve University	December 2022 – May 2024
Privacy-Preserving Sharing of Genomic Databases - Architected secure sandbox environment enabling privacy-preserving data sharing in genome-wide association studies (GWAS), supporting collaborative research while maintaining data privacy compliance with NIH Genomic Data Sharing policy - Engineered server-side pipelines for raw genomic data processing and metadata computation, accelerating collaborative analysis while reducing manual preprocessing overhead - Implemented privacy-preserving preview techniques for GWAS results, allowing collaborators to assess study feasibility prior to IRB approval, and enabling compliance with NIH Genomic Data Sharing policy - Implemented a storage architecture using MongoDB for genomic metadata - Created secure APIs, utilizing Flask.py and Express.js, facilitate secure sharing of genomic data across institutions - Implemented preliminary assessment tools for identifying and removing low-quality or biased data - Developed techniques for various data sharing scenarios, including statistics sharing and complete database sharing <i>Supported by NSF #2141622 and NIH 5R01LM014520-02</i>		

Halo Harbor: Privacy-Preserving Distributed Data Architecture

- Contributed to the design and development of an architecture enabling secure, user-controlled data sharing between personal data sources and service providers
- Worked under the direction of Professors Youngjin Yoo and Erman Ayday
- Leveraged DIDComm protocols, User Data Agents, Verifiable Credentials, and Hyperledger Aries Cloud Agent
- Enabled secure peer-to-peer communication with cryptographically secure methods that avoided centralized vulnerabilities
- Integrated modular “data plugs” to pull data from various sources, process it through a mapping layer, and enforce sharing rules via an access control layer
- Incorporated edge computation, a digital wallet for credentials, and a data repository with user-centric control mechanisms for temporary storage to ensure secure data handling

INDUSTRY EXPERIENCE

Full-Stack Engineer

bitHuman.ai

March 2025 – June 2025

- Operated in fast-paced, venture-backed AI startup environment reporting to CTO, contributing to strategic product decisions, delivering features and bug fixes while collaborating across design and marketing teams
- Developed multi-service visual AI agents platform with real-time conversational AI and lifelike representations, using Next.js 14+, React, TypeScript, Python (FastAPI), and Azure Functions across microservices
- Engineered real-time communication infrastructure using LiveKit WebRTC and Azure SignalR for low-latency multimodal voice and video interactions with sub-100ms response times
- Enhanced AI service orchestration layer with bug fixes and new API endpoints supporting multiple providers (OpenAI, Deepgram, Cartesia TTS, xAI) with configurable STT/TTS pipelines
- Developed Picture-in-Picture agent interface enabling users to minimize AI agents into floating windows with full interactive controls and cross-application state synchronization for adaptive usability
- Built imagineX agent management console with user-centric interfaces for agent persona, voice, and look customization, credit-based billing, organization user management, and analytics dashboards using React and ShadCN
- Position eliminated along with CTO and team during company restructuring, CEO available as reference

Software Development Engineer, Intern, Sunnyvale, CA

Amazon Lab126

May – August 2023

- Developed the Alexa Multi-Modal Media Player, enhancing the streaming experience for millions of Alexa users by reducing latency by 8% and optimizing performance
- Redesigned the Fire OS streaming algorithm to dynamically adjust bitrates based on internet latency, reducing buffering and improving playback efficiency
- Implemented a media fetching strategy that initially streams at the lowest available bitrate for early segments and differing audio formats before transitioning to higher bitrates
- Gained hands-on experience with Fire OS by working directly on media streaming modules and testing across Echo devices, including Echo Dot, Echo Show, and Echo Studio
- Streamlined testing processes by synchronously collecting and analyzing data from thousands of experiments, addressing challenges such as data loss from transfer cable limitations
- Practiced customer-centric engineering at Lab126, focusing on optimizing user experiences and delivering impactful improvements
- Technologies: C++, Fire OS, AWS, Media Codecs, and Media Streaming

Selected Projects

- *Health Data Platform for Progressive Insurance*
Developed privacy-preserving distributed architecture for secure Fitbit health data sharing using User Data Agent with ACA-Py and DIDComm protocols. Enabled user-controlled data flow and anonymized health insights through edge computation and multi-domain containerization on AWS ECS.
- *Urban Discovery System*
Built city matching service with a team of eight students for senior project, finding top 10 US cities based on user preferences by aggregating open-source data for weather, demographics, employment, and cost of living. TypeScript, Express.js, MongoDB, Python.